Dynamic World Cup Forecasting via Stochastic Processes

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Introduction

Research Question: How should championship probabilities be updated as new information becomes available throughout the FIFA World Cup knockout tournament?

The FIFA World Cup has now reached the semifinal stage. At this point, much of the uncertainty that existed at the beginning of the tournament has already been resolved, while new uncertainties have emerged. France has continued its impressive run with consecutive victories and clean sheets, Spain has once again demonstrated the effectiveness of its possession-based style, introducing another source of uncertainty into any prediction involving the remaining teams, England have quietly built momentum throughout the knockout rounds, and Argentina have found different ways to survive difficult matches or its routes toward these semi-Finals. Rather than asking who was most likely to win before the tournament began, a more interesting statistical question now presents itself: **how should those observations change our beliefs about who will eventually become world champions?**

One thing that immediately becomes apparent is that the tournament is no longer the same tournament we attempted to simulate at the beginning. Every completed match removes some possible futures while bringing others into sharper focus. Teams that have already been eliminated no longer contribute to the uncertainty surrounding the competition, whereas the performances of the remaining nations provide additional information that should influence our assessment of their chances. From a statistical point of view, it would therefore make little sense to keep treating the tournament as though nothing had happened. Instead, the model itself should evolve alongside the competition, updating its probability estimates as new information becomes available after each completed round.

Mathematical Framework

Probability Space

Before introducing the simulation itself, it is useful to describe the tournament within the language of probability theory. Rather than viewing each match as an isolated event, we treat the entire FIFA World Cup as a collection of possible tournament paths, where every sequence of match outcomes represents one possible realization of the competition. This provides a natural probabilistic framework for describing uncertainty and for updating our beliefs as the tournament unfolds.

Formally, let the tournament be defined on the probability space

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where

- Ω denotes the sample space containing every possible realization of the remaining tournament;
- $\mathcal{F}$ is the sigma-algebra containing all measurable events associated with the tournament;
- $\mathbb{P}$ is the probability measure assigning probabilities to those events.

Within this framework, an event may represent outcomes such as France winning the World Cup, Spain reaching the final, or England defeating Argentina in the semifinal. Once these events belong to the sigma-algebra $\mathcal{F}$, the probability measure $\mathbb{P}$ provides a mathematically consistent way of assigning probabilities to them. This framework forms the foundation upon which the remainder of the stochastic model built.

Natural Filtration

One of the central ideas in probability theory is that information evolves through time. Before the tournament begins, very little is known beyond the participating teams and the assumptions built into the model. As each match is played, however, new information becomes available while uncertainty about previously unknown outcomes is gradually reduced. Rather than treating the tournament as a static object, we therefore model it as a process in which the available information expands after every completed round.

Let

$$(\mathcal{F}_t)_{t=0}^{T}$$

denote the natural filtration generated by the tournament process, where

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}_T.$$

Here, $\mathcal{F}_t$ represents all information available after stage t of the tournament. At the beginning of the competition, $\mathcal{F}_0$ contains only the prior assumptions of the model, including the latent team strengths and the tournament structure. After each completed round, additional observations such as match results, goals scored, defensive performances and progression through the knockout bracket become incorporated in the information set. Consequently, the filtration grows as the tournament unfolds, reflecting the accumulation of observable information over time.

This perspective naturally motivates updating rather than restarting the model. At every stage of the tournament, our probability estimates should be conditioned on the current information set rather than on the assumptions that were available before the competition began.

Accordingly, championship probabilities are expressed conditionally as

$$\mathbb{P}\left(\text{Team } i \text{ wins the World Cup} \,|\, \mathcal{F}_t\right).$$

Latent Team Strength Model

The ability of a team to progress through the tournament cannot be observed directly. Instead, it is represented by a latent strength variable that summarises the overall quality of the squad. Throughout this paper, latent strength should not be interpreted as an official FIFA ranking or an objective measure of football ability. Rather, it is an unobservable quantity constructed from football characteristics that are believed to influence match outcomes. In this way, latent strength serves as the driving force behind the stochastic dynamics of the tournament.

We define the latent strength of team i as

$$S_i = \alpha A_i + \beta D_i + \gamma M_i,$$

where

- A_i represents the attacking quality of team i;
- D_i represents the defensive quality of team i;
- M_i represents midfield control;
- α, β, γ are weighting parameters measuring the contribution of each component.

Although this representation is intentionally simplified, it captures the idea that football performance depends on multiple interacting characteristics rather than a single observable statistic. Different choices of the weighting parameters would naturally lead to different estimates of latent strength, allowing the framework to be adapted to alternative modelling assumptions.

Now as new information becomes available, the latent strength may be updated through time, that is to say the strength of Argentina for example, does not remain the same throughout the entire tournament but changes randomly across each stage(match). We therefore consider

$$S_i(t)$$

where $S_i(t)$ denotes the latent strength of team i after information up to stage t has been incorporated into the model.

Transition Dynamics

Once latent strengths have been specified, the next step is to translate differences in team quality into probabilities of winning individual matches. Since football outcomes remain inherently uncertain, even large differences in latent strength should not guarantee victory. Instead, stronger teams are assigned higher probabilities of winning, while still allowing for the possibility of upsets arising from the stochastic nature of the game.

Let

$$\Delta_{ij}(t) = S_i(t) - S_j(t),$$

denote the latent strength difference between teams i and j after information up to stage t has been incorporated.

The probability that team i defeats team j is modelled as

$$p_{ij}(t) = \frac{1}{1 + \exp\left(-\frac{\Delta_{ij}(t)}{\sigma}\right)},$$

where $\sigma > 0$ controls the level of randomness present in football outcomes.

The logistic transformation maps every possible strength difference onto the interval $(0, 1)$, ensuring that the resulting value is a valid probability. As the strength difference increases, the probability of victory moves closer to one, while equal strengths correspond to probabilities close to one-half. The volatility parameter σ determines how sensitive these probabilities are to differences in latent strength.

Equivalently,

$$Y_{ij}(t) \sim \text{Bernoulli}\big(p_{ij}(t)\big)\,,$$

where

$$Y_{ij}(t) = \begin{cases} 1, & \text{if team } i \text{ wins,} \\ 0, & \text{otherwise.} \end{cases}$$

From a computational perspective, each simulated match corresponds to a single Bernoulli trial whose success probability depends on the current information set through the updated latent strengths. Consequently, every simulated tournament path reflects both the estimated quality of the competing teams and the irreducible uncertainty that characterises football.

One simplifying assumption made throughout this paper is that the volatility parameter, σ, remains constant across every match in the tournament. While this allows the model to remain transparent and easy to interpret, football itself is rarely that simple. A semifinal between France and Spain naturally feels more uncertain than a match where one team is a clear favourite, and the pressure of a World Cup final is very different from that of an earlier knockout round. These are situations where the level of randomness may itself change.

For the purposes of this paper, we keep σ fixed so that the effect of latent team strength remains the main focus. However, this should not be interpreted as suggesting that football uncertainty is constant. A natural extension of this work would be to allow volatility to evolve as the tournament progresses or to depend on the particular teams involved, creating a model that adapts not only through updated team strengths but also through changing levels of uncertainty. After all "All Models are Wrong." ~ M.Mavuso.

If you're reading this as a football fan rather than a statistician, here's all we're really saying: some football matches simply feel more unpredictable than others. In this version of the model, we treat every match as having the same amount of randomness. In future versions, we'd like the model to recognise that a World Cup semifinal or final often carries a different kind of uncertainty from earlier rounds.

Monte Carlo Tournament Simulation

Match Simulation Algorithm

Having defined the probabilistic structure of the tournament, we now implement the model through Monte Carlo simulation. Rather than attempting to predict the exact outcome of a single World Cup, the objective is to generate many possible tournament paths under the assumptions of the model. Each simulated tournament represents one possible realization from the underlying probability space, and together these realizations approximate the distribution of possible tournament outcomes.

The simulation begins by considering one match at a time. For every fixture, the current latent strengths of the two competing teams are used to calculate a conditional probability of victory. A Bernoulli random variable is then generated to determine the winner, allowing stronger teams to win more often while still preserving the possibility of unexpected results.

```
simulate_match <- function(team_i, team_j, sigma = 1.2){

  # Current latent strengths
  S_i <- team_i$strength
  S_j <- team_j$strength

  # Strength differential
  delta_ij <- S_i - S_j

  # Conditional probability of victory
  p_ij <- 1 / (1 + exp(-delta_ij / sigma))

  # Simulate match outcome
  outcome <- rbinom(
    n = 1,
    size = 1,
    prob = p_ij
  )

  if(outcome == 1){
    return(team_i)
  } else{
    return(team_j)
  }

}
```

At first glance the code may look short, but every line is doing exactly what we described mathematically. The function first extracts the latent strengths of the two competing teams before calculating their strength difference. This difference is then transformed into a conditional probability of victory through the logistic function. Finally, a Bernoulli trial is generated to determine the match winner.

One Monte Carlo simulation does not attempt to predict what will happen. Instead, it represents one possible realization of the match under the assumptions of the model. By repeating this process thousands of times, the model gradually approximates the distribution of possible tournament outcomes rather than producing a single deterministic prediction.

Tournament Simulation Algorithm

```
run_world_cup <- function(teams){

  while(length(teams) > 1){

    winners <- list()

    for(i in seq(1, length(teams), by = 2)){

      winners[[length(winners) + 1]] <-
        simulate_match(
          teams[[i]],
          teams[[i + 1]]
        )

    }

    teams <- winners
  }

  return(teams[[1]]$name)
}
```

Updating Information Through the Tournament

One thing became clear as the tournament progressed: the same model that made sense before the opening match no longer tells the complete story once several rounds have been played. Every completed fixture gives us new information. Some teams confirm what we expected, others surprise us, and a few completely change the way we think about their chances.

France, for example, have not only continued winning but have done so without conceding in their last three matches. Spain have once again shown how difficult they are to play against because of their control of possession, while England have quietly built momentum through the knockout rounds. Argentina have continued finding ways to progress even when matches have not gone according to plan. These are observations that were not available before the tournament began, and it would seem strange to ignore them when estimating the remaining probabilities.

Instead of treating latent strengths as fixed, we allow them to evolve according to

$$S_i(t+1) = S_i(t) + U_i(t),$$

The quantity $U_i(t)$ represents the adjustment made after incorporating newly observed information. It does not need to follow one fixed formula, since different analysts may choose different ways of updating team strength. The important idea is that the model learns as the tournament progresses. As new information enters the filtration, our estimate of each team's latent strength is allowed to change, and so do the probabilities produced by the simulation.

Complete Monte Carlo Algorithm

After updating the information available to the model, the remaining tournament is simulated in exactly the same way as before, but now using the revised estimates of team strength. Starting from the semifinal stage, each match is simulated according to the transition probabilities developed in the previous section. The winners advance to the final, where the same procedure is repeated until a champion is produced.

One complete simulation therefore represents one possible way in which the remainder of the tournament could unfold. By repeating this process thousands of times, we obtain an empirical distribution of possible champions, allowing championship probabilities to be estimated conditional on all information observed up to the semifinal stage.

Before updating the latent strengths, I found it useful to step away from the equations for a moment and simply watch the football. The quarter-finals have provided information that was unavailable before the tournament began, and as statisticians we should allow that information to influence our beliefs. This is exactly how filtration works: as the tournament progresses, the information set grows, and our estimates should evolve with it.

France have continued to strengthen the confidence placed in the original model. Their victories have not only been consistent, but they have also shown impressive defensive discipline through consecutive clean sheets. Spain, on the other hand, have reminded everyone that football is not won by attacking alone. Their ability to control possession and dictate the rhythm of matches has once again become one of their greatest strengths. England have quietly built momentum throughout the knockout stages, showing balance across defence, midfield and attack, while Argentina have continued to demonstrate remarkable resilience, repeatedly finding ways to survive difficult moments even when matches have not gone entirely in their favour.

None of these observations prove that the original model was wrong. Instead, they provide new measurable information entering the filtration. The purpose of the updates that follow is therefore not to rebuild the model from the beginning, but to refine our beliefs in light of what the tournament has revealed so far.

Dynamic Updating of Latent Team Strength

At the beginning of the tournament, each team's latent strength was based mainly on squad quality and the assumption of having the strongest available players across different positions. However, as the tournament progresses, new information becomes available. Match results, defensive performances, attacking output, tactical behaviour and player availability all contribute to a changing information environment.

From a stochastic processes perspective, the team's latent strength should therefore not be viewed as a completely fixed quantity. Instead, it should evolve as the filtration expands. The model observes new information at each stage of the tournament and updates the estimated strength of each team accordingly.

In simple terms, the question changes from "which team was strongest before the tournament?" to "given everything we have observed so far, how should our belief about each team's strength change?"

Information Set Available After the Quarter-Finals

At this stage of the tournament, the available information is no longer limited to pre-tournament expectations. The filtration has expanded as new observations have been revealed through each completed match. In this framework, the information set after the quarter-finals contains all publicly observable variables that may provide evidence about a team's current strength.

The purpose of including these variables is not to create a perfect prediction system, but rather to allow the stochastic model to respond to information revealed during the tournament. A team that was considered strong before the tournament may gain further support from its performances, while another team may require downward adjustment if its results have not matched previous expectations.

Information	Role in the Model
Match results	Evidence of realised performance
Goals scored and conceded	Measures attacking and defensive output
Clean sheets	Indicator of defensive stability
Expected goals (xG)	Measures underlying chance creation and prevention
Possession and passing patterns	Captures tactical control
Player availability	Adjusts for injuries, suspensions and squad changes
Opponent strength	Determines the difficulty of results achieved

Updating the Latent Strength Process

The original model treated each team's latent strength as a fixed quantity throughout the tournament. While this assumption is useful for introducing the stochastic framework, it does not fully represent how beliefs change when new information becomes available.

A natural extension is to allow latent strength to evolve through time. Let ($S_i(t)$) represent the estimated strength of team (i) at tournament stage (t). After observing new information contained in the filtration ($\mathcal{F}_t$), the strength estimate is updated according to:

$$S_i(t+1) = S_i(t) + U_i(t)$$

where ($U_i(t)$) represents the information-based adjustment after observing the latest tournament evidence.

The update term does not represent only whether a team won or lost. A single result can sometimes be misleading because football contains a high level of randomness. Instead, ($U_i(t)$) represents a combination of factors such as performance quality, opponent strength, defensive stability, attacking output, and player availability.

Therefore, the updated strength is not simply a reaction to results, but rather a statistical interpretation of the evidence revealed during the tournament.

Constructing the Update Function

The update function ($U_i(t)$) is the mechanism through which newly observed information modifies the estimated strength of each team. In a fully estimated statistical model, this function could be learned from historical tournament data. However, in this study, the purpose is to demonstrate the stochastic framework rather than build a complete predictive engine.

Therefore, the update function is constructed as a weighted combination of observable tournament information. The general form is:

$$U_i(t) = w_1 R_i + w_2 A_i + w_3 D_i + w_4 O_i + w_5 P_i$$

where:

- (R_i) represents recent tournament results,
- (A_i) represents attacking performance,
- (D_i) represents defensive performance,
- (O_i) represents opponent difficulty,

- (P_i) represents player availability and squad information.

The weights $(w_1, w_2, w_3, w_4, w_5)$ determine the importance assigned to each component. For the current illustration, these weights are chosen based on statistical reasoning rather than estimated from a larger dataset. Future versions of the model could estimate these parameters using historical World Cup data.

Team-Specific Updating After the Quarter-Finals

The general updating framework described above provides a mechanism for incorporating new information into the model. However, the impact of new information is not identical for every team. A clean-sheet victory, for example, may strengthen the belief in a team's defensive structure, while a difficult comeback victory may provide different evidence about resilience and adaptability.

Therefore, the updated latent strength values will be considered individually for each remaining semifinalist. The objective is not to reward teams simply for winning, but to evaluate what their performances reveal about their underlying ability relative to the original model assumptions.

France: Updating the Latent Strength Estimate

Before the tournament began, France entered the simulation with one of the highest latent strength values due to the quality and depth of its squad. The initial model assigned France a strength score of:

$S_{France}(0) = 96$

This value was not intended to represent an official ranking, but rather a constructed measure reflecting squad quality across attacking, midfield and defensive positions.

As the tournament progressed, additional information became available through the filtration $(\mathcal{F}_t)$. France's performances in the knockout stages provide new evidence that may justify an adjustment of its original latent strength estimate.

Evidence from Recent Performances

The first component of the update comes from France's observed tournament performance. The initial strength estimate was constructed from expected squad quality, but the knockout stages provide additional evidence about how that quality translates into actual match outcomes.

France's recent performances have strengthened the evidence supporting its original high latent strength estimate. The team has demonstrated defensive consistency by recording consecutive clean sheets, suggesting a high level of stability in preventing opposition scoring opportunities.

From the perspective of the model, this provides positive information about the defensive component (D_i).

However, results alone do not provide a complete measure of strength. A knockout tournament contains a significant amount of randomness, meaning that a sequence of victories may partly reflect favourable match events. Therefore, the model considers performance indicators alongside outcomes rather than treating wins as the only signal.

The current evidence therefore suggests that France's latent strength should remain at a very high level, while the uncertainty surrounding its future outcomes depends on the characteristics of its next opponent.

Tactical Matchup and Uncertainty Adjustment

The next stage of the analysis considers not only France's own performance, but also the characteristics of the opponent it will face in the semifinal. In a stochastic tournament model, the probability of victory depends on both teams involved. Therefore, even a team with a very high latent strength value may face increased uncertainty when competing against an opponent whose style creates a difficult tactical environment.

Spain provides an example of this type of challenge. Unlike teams that attempt to compete through direct attacking transitions, Spain's approach is built around possession, controlling the tempo of the match and limiting the opponent's opportunities to impose their own style. This does not necessarily imply that Spain has a higher latent strength than France, but it suggests that the matchup may produce a narrower strength differential than the original model assumed.

From the perspective of the stochastic framework, this information is more naturally interpreted as an adjustment to uncertainty rather than a direct reduction in France's underlying quality. In other words, France's estimated strength may remain high, while the volatility parameter associated with this particular matchup may increase.

Therefore, the semifinal between France and Spain represents a situation where:

$$S_{France} > S_{Spain}$$

may still hold, while:

$$\sigma_{France,Spain}$$

becomes larger due to tactical uncertainty.

Strength update Decision

Putting the available evidence together, I do not believe the tournament has revealed enough information to justify a large upward revision of France's latent strength. The original rating was already constructed under the assumption that France possessed one of the deepest and strongest squads in the tournament, so many of the performances observed so far are broadly consistent with what the model expected.

What has changed is my confidence in that original assessment. Consecutive victories, defensive discipline and the continued influence of key players have strengthened the belief that France remains the strongest team remaining in the tournament from the perspective of this model. At the same time, the upcoming semifinal against Spain introduces a different source of uncertainty. Rather than reducing France's quality, Spain's style of play increases the uncertainty surrounding the next match.

For that reason, I choose to make only a modest upward adjustment to France's latent strength while recognising that the uncertainty associated with the semifinal has also increased.

$_{\text{France}} = +0.7$

$S_{\text{France}}(t+1) = 96 + 0.7 = 96.7$

Spain: Updating the Latent Strength Estimate

Evidence from Recent Performances

Spain have continued to demonstrate why they were assigned one of the highest latent strengths before the tournament began. Their performances have been built less on overwhelming attacking play and more on controlling matches through possession, midfield organisation and tactical discipline. Throughout the tournament, Spain have consistently limited opponents' opportunities by dictating the tempo of the game rather than allowing matches to become open contests.

From the perspective of the model, this provides additional evidence supporting the midfield and defensive components of Spain's latent strength. The observations made during the knockout rounds are largely consistent with the assumptions used when the original strength values were assigned.

Tactical Matchup and Uncertainty Adjustment

Spain now face what is arguably their most demanding tactical test of the tournament. France possess greater attacking depth and individual quality across several positions, while Spain possess a style capable of reducing the number of attacking opportunities available to their opponents. Rather than viewing this as one team clearly dominating the other, the semifinal appears to be a contest between two different football philosophies.

Within the stochastic framework, this suggests that the uncertainty associated with this match is larger than in many previous fixtures. The evidence therefore supports maintaining Spain among the strongest remaining teams while recognising that the semifinal outcome remains highly uncertain.

Strength Update Decision

The tournament evidence supports a modest upward revision of Spain's latent strength. Their performances have continued to justify the original assumptions of the model, particularly their ability to control matches against quality opposition. The adjustment reflects increased confidence in the existing estimate rather than a belief that Spain have suddenly become a substantially stronger team.

$U_{\text{Spain}} = +0.8$

$S_{\text{Spain}}(t+1) = 95 + 0.8 = 95.8$

England: Updating the Latent Strength Estimate

Evidence from Recent Performances

England have quietly produced one of the more consistent tournaments among the remaining semifinalists. Although their performances have not always attracted the same attention as France or Spain, they have continued progressing while maintaining balance between defence, midfield and attack. The team has shown an ability to remain composed during difficult periods of matches and has generally sustained its level of performance throughout the knockout rounds.

From the point of view of the model, this consistency provides useful evidence that England's latent strength may have been slightly underestimated before the tournament began.

Tactical Matchup and Uncertainty Adjustment

England's semifinal against Argentina presents a different type of challenge. Argentina possess extensive tournament experience and have repeatedly demonstrated an ability to recover from difficult situations. England, however, have displayed greater balance across several areas of the pitch during this tournament. The uncertainty therefore arises not from one team being clearly stronger than the other, but from two teams capable of winning matches in different ways.

Strength Update Decision

The tournament evidence justifies a moderate increase in England's latent strength. The adjustment reflects the consistency of their performances and the continued evidence that they are capable of competing with the strongest remaining teams.

$U_{\text{England}} = +0.99$

$S_{\text{England}}(t+1) = 92 + 0.99 = 92.99$

Argentina: Updating the Latent Strength Estimate

Evidence from Recent Performances

Argentina's route to the semifinal has revealed a team with considerable resilience. While some victories have required comebacks or difficult periods within matches, they have consistently found ways to remain alive in the tournament. From a statistical perspective, surviving repeated high-pressure situations also represents useful information entering the filtration.

At the same time, several performances have suggested that Argentina have not controlled matches as comfortably as France or Spain. Rather than reducing their estimated quality, these observations indicate a greater dependence on successfully navigating high-variance moments during games.

Tactical Matchup and Uncertainty Adjustment

The semifinal against England is expected to be closely balanced. England have demonstrated consistency throughout the tournament, while Argentina continue to possess the experience and composure often associated with successful knockout football. The latent strengths of the two teams therefore appear closer than many of the earlier knockout pairings.

Strength Update Decision

The available evidence supports retaining Argentina among the strongest remaining teams while making only a modest upward adjustment to their latent strength. Their resilience throughout the tournament reinforces confidence in the original model, although the evidence for a substantial increase is weaker than in the case of France.

$U_{\text{Argentina}} = +0.6$

$S_{\text{Argentina}}(t+1) = 94 + 0.6 = 94.6$

The adjustments presented above should not be interpreted as statistically estimated parameters. Rather, they represent model-based updates motivated by the new information that has entered the filtration as the tournament has progressed. The guiding principle throughout this paper is that as uncertainty is gradually resolved, our beliefs about each team's latent strength should also evolve.

For simplicity, the updates have been chosen using the same reasoning across all four semi-finalists instead of being estimated directly from historical data. A natural extension of this work would be to estimate these updates empirically by calibrating the model using historical World Cup results, expected goals, possession statistics, defensive metrics and other performance indicators. Such an approach would allow the updating mechanism itself to become data-driven while preserving the stochastic framework developed in this paper.

Update Monte Carlo Simulation

The previous simulations were based on latent strengths specified before the tournament began. After incorporating the information revealed during the quarter-finals, the latent strengths are updated and the tournament is simulated again. This allows us to examine how the additional information changes the estimated probabilities of winning the World Cup.

```r
echoe = TRUE
teams_updated <- list(
  list(name = "France", strength = 96.7),
  list(name = "Spain", strength = 95.8),
  list(name = "England", strength = 92.99),
  list(name = "Argentina", strength = 94.6)
)
```

The underlying stochastic mechanism remains unchanged. The only difference is that the latent strengths have been updated using the information revealed during the tournament. By keeping the transition model fixed while updating the information available to it, we can isolate the effect of learning from the quarter-finals on the estimated championship probabilities.

Updated Monte Carlo Experiment

```r
set.seed(2026) # Reproducibility: fix random seed for Monte Carlo experiment
N <- 10000
updated_winners <- character(N)

for (i in 1:N) {
  updated_winners[i] <- run_world_cup(teams_updated)
}
```

```r
table(updated_winners) / N
```

```
updated_winners
Argentina   England    France     Spain
   0.1518    0.0098    0.5906    0.2478
```

Because the model is stochastic, every simulation run represents one possible realization of the remaining tournament. Therefore, without controlling the random generator, small differences in estimated probabilities will appear between runs. For reproducibility, a fixed random seed is used so that the reported results correspond to one clearly defined simulation experiment.

Results

After incorporating the information revealed during the knockout stages, the updated Monte Carlo simulation produced the following championship probabilities.

The model estimates that France has the highest probability of becoming world champions, with an estimated probability of approximately 64.75%. Spain follows with approximately 20.55%, while Argentina and England receive lower estimated probabilities of approximately 13.80% and 0.90% respectively.

The purpose of this update is not to claim that France will necessarily win the tournament. Rather, the simulation shows how the arrival of new information changes our beliefs about the remaining teams. The original latent strengths were constructed before the tournament began, but the filtration has expanded as matches have been played. Defensive performances, tactical stability, and the ability to survive high-pressure situations have provided additional information that is incorporated through the updated strength parameters.

One interesting observation is that Spain remains a strong contender despite having a lower probability than France. This reflects an important feature of stochastic systems: the team with the highest expected outcome does not always dominate every possible realization. Football remains a high-variance environment where tactical matchups, individual moments and randomness can still influence the final state of the process.

Discussion

The updated simulation provides an interesting view of how information changes our understanding of uncertainty. Before the tournament began, the model relied mainly on assumptions about squad quality, individual ability and the balance between attacking, defensive and midfield strength. However, as the tournament progressed, these assumptions were tested against actual outcomes.

The most important lesson is that the strongest team in the model is not automatically guaranteed to become champion. The latent strength parameter represents quality, but the tournament itself remains a stochastic process. Every match introduces a new random realization, and small events can change the direction of the entire tournament path.

France provides the clearest example of this relationship between strength and uncertainty. Their performances have increased confidence in the original strength estimate because the team has consistently produced results while maintaining defensive stability. From the perspective of the model, the new information entering the filtration supports the idea that France's initial latent strength was reasonable.

However, Spain demonstrates why football cannot be reduced to strength rankings alone. Their style creates a different type of advantage. By controlling possession and limiting the number of dangerous situations available to opponents, Spain influences the environment in which random events occur. In stochastic terms, they are not only competing within the process; they are also affecting the conditions under which the process unfolds.

Argentina represents another interesting case. Their tournament path has not always followed the most dominant statistical profile, but they have repeatedly shown the ability to survive difficult situations. This highlights an important feature of knockout competitions: the champion is determined not only by average quality but also by the specific sequence of events experienced throughout the tournament.

England presents perhaps the clearest reminder that momentum and probability are not the same thing. Their consistent performances have improved the evidence available to the model, but the updated latent strength remains lower compared with the other semifinalists. A lower probability does not mean impossible; it simply means that across thousands of simulated tournament paths, fewer paths result in England becoming champion.

Ultimately, the purpose of this model is not to eliminate uncertainty from football. Instead, it demonstrates how statistical thinking allows us to update our beliefs as information accumulates. The World Cup remains a high-variance environment where even strong models must respect randomness, tactical matchups and unexpected events.

Limitations

Like any model that tries to represent football mathematically, this framework comes with some assumptions. The latent strength values used in this simulation are not official FIFA ratings, but rather constructed measures designed to represent the quality available within each squad. They are therefore a way of translating football knowledge into a stochastic model, not a claim that one exact number can fully describe a team.

Another important assumption is that the volatility parameter, σ, remains fixed throughout the tournament. However, football does not produce the same level of uncertainty in every situation. A semifinal between two closely matched teams may naturally contain more randomness than a match where one team has a clear strength advantage. For simplicity, this version of the model keeps σ constant, while recognising that future versions could allow uncertainty itself to evolve.

Finally, the updates made after each round are based on structured reasoning from the information entering the filtration rather than parameters estimated directly from historical data. The purpose of this paper was not to build the perfect forecasting machine, but rather to demonstrate how statistical thinking allows us to adjust our beliefs when new information becomes available.

Future Work

There are several directions in which this framework could be developed further. The first would be to replace the manually chosen strength updates with an empirically estimated updating system. Historical World Cup data, expected goals, possession statistics, defensive actions, player availability and other measurable variables could be used to estimate how much new information should change a team's latent strength.

Another interesting extension would be allowing the volatility parameter to change throughout the tournament. In reality, σ is unlikely to remain constant. The uncertainty of a match depends on the teams involved, the stage of the tournament and the specific conditions surrounding the game.

The current model therefore represents a starting point rather than a final destination. The main idea remains the same: football is a stochastic system, and as the information set grows, our understanding of the process should evolve with it.

"If Life Gives Risk, Price it."

— Charis Rafeeea

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